Fiscal Year:	FY 2012	**Task Last Updated:**	FY 12/30/2011
PI Name:	Bailey, Susan M. Ph.D.		
Project Title:	Telomeric proteins in the radiation/DNA damage response		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	susan.bailey@colostate.edu	**Fax:**	FY 970-491-7742
PI Organization Type:	UNIVERSITY	**Phone:**	970-491-2944
Organization Name:	Colorado State University		
PI Address 1:	Environmental and Radiological Health Sciences		
PI Address 2:	1618 Campus Delivery		
PI Web Page:			
City:	Fort Collins	**State:**	CO
Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	01/01/2008	**End Date:**	12/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is now 12/31/2013 per NSSC information (Ed., 9/27/2012) NOTE: End date is now 12/31/2012 per NSSC information (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liber, Howard (Colorado State University)		
Grant/Contract No.:	NNX08AB65G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research addresses Radiation Health Bioastronautics Roadmap (BR) Carcinogenesis Questions 28c, d and j, and is aimed at advancing current knowledge and mechanistic understanding of the DNA damage processing that underlies radiation effects in general, and space radiation in particular. The goal of the studies proposed here is to provide information that will aid identification of susceptible individuals, ultimately improving risk prediction for radiation carcinogenesis in astronauts exposed to high energy charged particles in space. Summary: Appropriate cellular response to natural chromosomal termini (telomeres) and to exogenously created double-strand breaks (DSBs) is critical to maintaining genomic stability. Incorrect handling of DSB ends leads to mutation, chromosomal rearrangements, instability and carcinogenesis. Altered levels of telomeric proteins have been reported in some tumors. The interplay between telomeres and DSBs may be better understood not only by viewing the striking differences between them, but perhaps even more importantly by viewing the obvious similarity-they are both DNA ends, the very substrate both telomere and damage response/repair proteins specifically recognize and bind. We have been investigating the role DNA repair proteins play at telomeric ends. Recent and intriguing studies suggesting that the classic telomere-binding protein TRF2 is also found (rapidly and transiently) at sites of photo-induced DNA DSBs, has turned the question around-what are telomere proteins doing at DSB ends? Here, we propose to address this fascinating question by investigating the role(s) telomeric proteins play in ionizing radiation (IR)-induced DNA damage response/processing/repair. Our approach assesses mutation frequency and spectra at the molecular level, as well as extends the analyses to the cytogenetic level. Our preliminary studies lend support to the view that telomeric proteins aren't just for telomeres anymore. Increasing our understanding of previously unrecognized relationships between the radiation/DNA damage response and telomere function will facilitate predicting the carcinogenic risk of exposure to radiation in general, and to astronauts exposed to HZE particles in particular. Ultimately, we foresee utilizing telomere instability/dysfunction as a useful biomarker of underlying repair deficiency in order to identify individual susceptibility to radiation carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our investigation of interdependencies between DNA repair and telomeres will ultimately lead to identification and characterization of previously unappreciated telomere dysfunction, which in turn will provide a useful indication of an underlying repair defect. Therefore, telomere instability may ultimately serve as a sensitive biomarker, identifying individual susceptibility to radiation carcinogenesis in general and individual susceptibility to HZE exposure in particular.
Task Progress:	Appropriate cellular response to natural chromosomal termini (telomeres) and to exogenously created double-strand breaks (DSBs) is critical to maintaining genomic stability. Incorrect handling of DSB ends leads to mutation, chromosomal rearrangements, instability and carcinogenesis. Altered levels of telomeric proteins have been reported in some tumors. The interplay between telomeres and DSBs may be better understood not only by viewing the striking differences between them, but perhaps even more importantly by viewing the obvious similarity – they are both DNA ends, the very substrate both telomere and damage response/repair proteins specifically recognize and bind. We have been investigating the role DNA repair proteins play at telomeric ends (1,2). Studies suggesting that the “classic” telomere-binding protein TRF2 is also found (rapidly and transiently) at sites of photo-induced DNA DSBs, turned the question around – what are telomere proteins doing at DSB ends? We have recently shown that ionizing radiation (IR)-induced DSBs are insufficient to initiate TRF2 recruitment to damage sites (3). We are now working to further address this fascinating question by investigating the role(s) telomeric proteins play in IR-induced DNA damage response/processing/repair. Our overall aim is to study the effects of depleting expression of key telomeric binding proteins on DNA damage response/repair. The telomeric DNA-binding proteins TRF1 and TRF2 have multiple interacting binding partners, including TIN2, POT1 and tankyrase1. Many interactions between these players (and others) have been reported. Telomerase (hTERT) is also known to interact with many telomeric, as well as non-telomeric proteins (including DNA repair proteins), and extra-curricular/pro-tumorigenic activities of telomerase independent of its classical role in telomere length maintenance - including enhanced repair - have been proposed, but remains an open issue. Our approach has been to knockdown expression of these critical telomeric proteins in human cell lines utilizing siRNA strategies. DNA damage is induced by exposure to ionizing radiations (low LET gamma rays or HZE 56Fe ions). We then employ a powerful combination of mutational analyses and cytogenetic approaches to examine the phenotypic and genotypic consequences of the telomeric deficiencies on DNA repair endpoints. Experiments revealed that small interfering RNA (siRNA) knockdown of TRF2 results in a significant increase in radiation-induced mutation at the heterozygous thymidine kinase (tk) locus in human lymphoblastoid cells. Depletion of the Protection of Telomeres (POT1) protein also results in increased radiation mutagenesis. Both TRF2 and POT1 knockdowns behaved similarly, qualitatively and quantitatively; each led to elevated background mutant fractions, and also to enhanced radiation mutagenesis. These experiments were very recently repeated (Fall 2011 NSRL run); analyses are in progress. To further investigate the dynamics of the seemingly contradictory yet integrated cellular responses to the requisites of preserving telomere integrity while also efficiently repairing damaged DNA, we have also been investigating roles of the telomere associated Poly ADP ribose polymerase (PARP), tankyrase 1, in both telomere function and the DNA damage response following exposure to ionizing radiation. Tankyrase 1 siRNA knockdown in human cells significantly elevated recombination specifically within telomeres; i.e., telomere sister chromatid exchange (T-SCE), a phenotype with the potential of accelerating cellular senescence, and therefore aging (4). Most surprising however, was that depletion of tankyrase 1 resulted in concomitant and rapid reduction of the non-homologous end-joining protein DNA-PKcs, while Ku86 and ATM protein levels remained unchanged; DNA-PKcs mRNA levels were also unaffected. We found that the requirement of tankyrase 1 for DNA-PKcs protein stability reflected the necessity of its PARP enzymatic activity. We also demonstrated that depletion of tankyrase 1 resulted in proteasome-mediated DNA-PKcs degradation, explaining the associated defective damage response observed; i.e., increased sensitivity to ionizing radiation-induced cell killing, mutagenesis, chromosome aberration and telomere fusion. We provided the first evidence for regulation of DNA-PKcs by tankyrase 1 PARP activity and taken together, identified roles of tankyrase 1 with implications not only for DNA repair and telomere biology, but also for cancer and aging. We also presented our work at the Gordon Research Conference: Mammalian DNA Repair. The following articles have resulted from this work:

• Regulatory roles of tankyrase 1 at telomeres and in DNA repair: suppression of T-SCE and stabilization of DNA-PKcs Dregalla RC, Zhou J, Idate RR, Battaglia CL, Liber HL, Bailey SM. Aging (Albany NY) 2(10); Oct. 2010.

An invited News and Views article highlighting these findings, and putting them in perspective with our recent WRN/BLM/ERCC1 PNAS paper also resulted:

• The telomere protein tankyrase 1 regulates DNA damage responses at telomeres. Sandy Chang, Dept. of Laboratory Medicine, Yale University School of Medicine.

• Research Perspective: Telomere sister chromatid exchange and the process of aging. Blagoev KB, Goodwin EH and Bailey SM. Aging (Albany NY) 2(10); Oct. 2010.

Our results suggested that PARsylation may represent an important mechanism for regulating DNA-PKcs kinase activity, prompting us to consider whether pADPr modification influences DNA-PK's function in DNA repair. Therefore, we investigated the role of tankyrase dependent PARsylation of DNA-PKcs in NHEJ-mediated double-strand break (DSB) repair in human cells. We found that tankyrase PARP catalytic activity facilitated the recruitment to and function of DNA-PKcs at DSB sites, and in so doing provided the first insights into the relevance of regulation of DNA-PKcs by pADPr modification.

The following article has been under review at Nature Scientific Reports:

• Tankyrase 1-dependent pADPr modification of DNA-PKcs facilitates holoenzyme formation and function. RC Dregalla, DG Maranon and SM Bailey (2011).

Our studies continue to support the view that telomeres and DNA repair are intimately connected. Therefore, individual susceptibility factors involving both telomeres and DNA repair must be considered when estimating cancer risk following ionizing radiation exposures.

References

1. Williams, E.S., Klingler, R., Ponnaiya, B., Hardt, T., Schrock, E., Lees-Miller, S.P., Meek, K., Ullrich, R.L. and Bailey, S.M. (2009) Telomere dysfunction and DNA-PKcs deficiency: characterization and consequence. Cancer Res, 69, 2100-2107.

2. Fabre, K.M., Ramaiah, L., Dregalla, R.C., Desaintes, C., Weil, M.M., Bailey, S.M. and Ullrich, R.L. (2011) Murine Prkdc polymorphisms impact DNA-PKcs function. Radiation research, 175, 493-500.

3. Williams, E.S., Stap, J., Essers, J., Ponnaiya, B., Luijsterburg, M.S., Krawczyk, P.M., Ullrich, R.L., Aten, J.A. and Bailey, S.M. (2007) DNA double-strand breaks are not sufficient to initiate recruitment of TRF2. Nat Genet, 39, 696-698; author reply 698-699.

4. Hagelstrom, R.T., Blagoev, K.B., Niedernhofer, L.J., Goodwin, E.H. and Bailey, S.M. (2010) Hyper telomere recombination accelerates replicative senescence and may promote premature aging. Proc Natl Acad Sci U S A, 107, 15768-15773.

Bibliography Type:	Description: (Last Updated: 12/10/2025)
Abstracts for Journals and Proceedings	Sishc B, Eadim A, Maranon DG, Goodwin EH, Bailey SM. "Influence of radiation quality and genomic location on sister chromatid exchange frequencies." Presented at 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. Abstract Book. p. 127-128. , Aug-2011
Abstracts for Journals and Proceedings	Dregalla RC, Maranon DG, Liber HL, Bailey SM. "Tankyrase 1-dependent pADPr modification of DNA-PKcs facilitates holoenzyme formation and function." Presented at 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. Abstract Book. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7037.pdf , Sep-2011
Articles in Peer-reviewed Journals	Fabre KM, Ramaiah L, Dregalla RC, Desaintes C, Weil MM, Bailey SM, Ullrich RL. "Murine Prkdc polymorphisms impact DNA-PKcs function." Radiat Res. 2011 Apr;175(4):493-500. Epub 2011 Jan 25. PMID: 21265624 , Apr-2011
Books/Book Chapters	Williams ES, Cornforth MN, Goodwin EH, Bailey SM. "CO-FISH, COD-FISH, ReD-FISH, SKY-FISH." in "Telomeres and Telomerase: Methods and Protocols. Second edition." Ed. Z. Songyang. Totowa, N.J. : Humana/Springer, 2011. p. 113-124. (Methods in molecular biology ; v. 735) ; http://dx.doi.org/10.1007/978-1-61779-092-8_11 ; also appears in PubMed: PMID: 21461816 , Apr-2011